

Project Short Description & Repository

This repository contains the Production Inventory Management System (PIMS) — a full-stack application for manufacturing inventory, purchases, orders, sales and ML-powered demand forecasting.

Track changes or contribute at: <https://github.com/arryaanjain/Orah.git>

Working Modules (what currently works)

- Authentication
 - Google OAuth sign-in via Laravel Socialite.
 - Sanctum token issuance and frontend callback handling.
 - Profile completion (company link / create) and company-scoped data isolation.
- Dashboard
 - Company-level summary metrics (products, orders, sales, low-stock).
 - Recent activity feed and inventory summary.
 - Shows ML alerts when analytics service is available; graceful fallback when not.
- Raw Material Master (RM)
 - CRUD for raw materials.
 - Multi-unit conversion support (base unit + conversion factors).
 - Frontend: responsive grid, modal forms, unit management.
- Raw Material Purchase
 - Single and batch purchase entry (supplier, batch, expiry, totals).
 - Cascading dropdowns: material → units.
 - Purchase history table and mobile-friendly card view.
- Products & BOM
 - Finished products CRUD, base unit, pricing, active flag.
 - Product Bill of Materials (BOM) to map required raw materials.
- Orders & Sales
 - Order creation and status workflow.
 - Sales recording, payment status, link to orders.
 - Dashboard aggregation for top products and sales overview.
- Customers
 - Customer CRUD and contact management.
- Analytics (ML)
 - Separate FastAPI service (ARIMA/SARIMAX models) providing:
 - /forecast (per warehouse + category)
 - /reorder-alerts
 - Laravel acts as a proxy; frontend shows analytics or "analytics unavailable" if ML service down.
 - 53 trained models saved; endpoints return confidence intervals and suggested reorder amounts.

Future Scope (priority ideas)

Short term

- Inventory valuation methods (FIFO, LIFO, weighted average) and cost reporting.
- Role-based access control (admin, purchaser, sales, viewer).
- CSV import/export for materials, purchases, products, customers.
- More robust UI testing and E2E tests.

Medium term

- Improve forecasting:
 - Replace/augment ARIMA with LightGBM or LSTM ensembles for complex seasonality.
 - Add automated retraining and model monitoring (drift detection).
- MRP-style production planning combining BOM + forecasts.
- Reorder automation with supplier lead times and safety stock optimization.
- Advanced reports (aging, lead-time, procurement spend).

Long term / integrations

- Integrate with accounting / ERP (QuickBooks, SAP) and e-invoicing.
- Warehouse optimization (routing, multi-warehouse allocation).
- Mobile apps (React Native) for scanning, stock adjustments, and approvals.
- Containerize (Docker) and CI/CD pipeline for easier deployment.
- Multi-region deployment and high-availability ML serving.

Contributions

- Contributions, issues and feature requests welcome at: <https://github.com/arryaanjain/Orah.git>
- Please fork, open PRs, and reference issues for larger changes.